

Safai Market - Phase 1 to Phase 4 Implementation Report

Annapurna Traders - Smart Billing for Retail Shops

Report generated: May 30, 2026

Executive Summary: Phases 1 through 4 have been fully implemented. Phase 1 fixed 18 critical bugs covering security, data integrity, and billing correctness. Phase 2 added complete GST billing (HSN codes, CGST/SGST/IGST, tax-invoice receipts). Phase 3 introduced Device Center, Bill Settings page, and Multi-Unit Variants UI. Phase 4 added Framer Motion page transitions, Web Audio API sound effects, cart badge animations, and CSS micro-interactions.

1. Phase 1 Summary - Critical Bug Fixes

Phase 1 addressed 18 documented bugs spanning security, data integrity, and billing correctness.

			Bug ID
BUG-001	Stock race condition	SELECT FOR UPDATE prevents concurrent stock depletion	
BUG-002	Shop isolation	GET /bills/:id and CANCEL now filter by shopId	
BUG-003	Walk-in customer	0 and null both mean Walk-in (no udhaar without real customer)	
BUG-004	Profit accuracy	Bill-level discount now subtracts from estimated profit	
BUG-005	Bill number collision	Crypto random suffix prevents concurrent duplicates	
BUG-006	Query performance	Status + categoryId pushed into SQL WHERE instead of JS filtering	
BUG-007	Shop isolation	GET /bills list filters by shopId at SQL level	
BUG-008	Stock race condition	SELECT FOR UPDATE for all bill operations	
BUG-009	Data isolation	shop_id added to activity_log; passed on insert	
BUG-010	Cart persistence	Cart localStorage key now includes shopId	
BUG-012	Infinite redirect	/auth/onboarding added to PUBLIC_PATHS	
BUG-013	Settings sync	syncFromShop() called on auth load; persistToServer() on save	
BUG-016	Discount validation	Negative discounts rejected; clamped to [0, unitPrice]	
BUG-017	UI overflow	Cart badge safe-area padding; checkout on top of footer	
BUG-018	Walk-in handling	0/null Walk-in handling	
BUG-019	Onboarding redirect	Explicit setLocation("/") after shop creation	
BUG-020	Bundle stock	Real stock check for each bundle item	
BUG-021	Keyboard scanner	Keyboard wedge scanner works on billing screen	

Files Modified (Phase 1)

lib/db/migrations/001_gst_and_fixes.sql (NEW)
lib/db/src/schema/products.ts
lib/db/src/schema/activity_log.ts
artifacts/api-server/src/routes/bills.ts
artifacts/api-server/src/routes/products.ts
artifacts/safai-market/src/components/auth-guard.tsx
artifacts/safai-market/src/components/auth-provider.tsx
artifacts/safai-market/src/pages/auth/onboarding.tsx
artifacts/safai-market/src/pages/billing/index.tsx
artifacts/safai-market/src/stores/cart.ts
artifacts/safai-market/src/stores/settings.ts
artifacts/safai-market/src/lib/receipt.ts
artifacts/safai-market/src/pages/settings/store.tsx

Migration Changes

- Added hsn_code, gst_rate, gst_inclusive to products table
- Added shop_id to activity_log table
- Added cancel_reason to bills table

Created 6 performance indexes (products, bills, customers, activity_log, bill_items)

2. Phase 2 Summary - GST Billing

HSN & GST Support

- HSN Code field added to product form (new + edit)
- GST Rate field (0%, 5%, 12%, 18%, 28%) with dropdown
- GST Inclusive / Exclusive toggle with live preview
- CGST/SGST split shown in real-time (rate/2 each)

Tax Invoice Receipt

- Receipt shows "TAX INVOICE" when GST enabled, "RETAIL BILL" otherwise
- Store GSTIN printed in header
- HSN code shown under each item line
- CGST, SGST (or IGST for inter-state) shown in totals table
- Taxable amount calculated and displayed separately

Cart GST Breakdown

- `getGstBreakdown()` method in cart store computes CGST/SGST/IGST
- Checkout sheet displays GST totals before confirming bill
- GST fields (`gstRate`, `gstInclusive`) carried on every `CartItem`

Files Modified (Phase 2)

```
lib/db/src/schema/products.ts (GST columns)
artifacts/safai-market/src/pages/products/new.tsx (GST form fields)
artifacts/safai-market/src/pages/products/edit.tsx (GST form fields)
artifacts/safai-market/src/stores/cart.ts (getGstBreakdown, GST fields)
artifacts/safai-market/src/pages/billing/index.tsx (GST checkout display)
artifacts/safai-market/src/lib/receipt.ts (Tax Invoice format)
artifacts/safai-market/src/pages/settings/store.tsx (GST settings fields)
```

3. Phase 3 Summary - Device Center, Bill Settings, Multi-Unit Variants

3.1 Device Center

- Scanner list: USB Barcode Scanner, Bluetooth Scanner, Camera Scanner
- Printer list: Thermal Printer, Browser Print (PDF)
- Test Scanner button - shows "Waiting for scan..." state
- Test Print button - opens browser print dialog with test page
- Keyboard Wedge Mode toggle
- Bluetooth pairing instructions and thermal printer setup guide

3.2 Bill Settings Page

- Paper size selector (58mm / A4 / A5)
- Footer message editor
- Receipt toggles: Discount Amount, GST Breakdown, Estimated Profit
- Live receipt preview (print dialog opens with sample bill)
- Sound Effects and Animations toggles

3.3 Multi-Unit Variants

- Lists existing variants (child products by parentProductId)
- Quick-size chips: 100g, 250g, 500g, 1kg, 100ml, 1L, Small, Medium, Large
- Per-variant: name, sell price, buy price, stock, barcode
- Creates separate product records in DB linked via parentProductId

3.4 Routes & Menu

- 3 new routes added to App.tsx
- 2 new items added to More > Settings section
- "Variants" button added to Product Detail header

Files Created (Phase 3)

```
src/pages/settings/devices.tsx
src/pages/settings/bill-settings.tsx
src/pages/products/variants.tsx
```

Files Modified (Phase 3)

```
src/App.tsx (3 new routes)
src/pages/more/index.tsx (menu items)
src/pages/products/detail.tsx (Variants button)
```

4. Phase 4 Summary - Animations, Micro-interactions & Sound Effects

4.1 Animation System

- pageVariants - fade in + slide up (0.18s) on every route change
- successVariants - spring scale-in for Bill Success checkmark
- staggerContainer + staggerItem - buttons stagger in after bill success
- cartBadgeVariants - spring bounce when item count changes
- scaleInVariants - for dialog/modal entrances

4.2 Sound Effects (Web Audio API - zero dependencies)

Event	Sound	Trigger
Cart Add	Double beep (880Hz + 1100Hz)	Manual tap or keyboard scan
Scan Success	High beep (1320Hz + 1760Hz)	Barcode scanner (USB BT camera)
Bill Success	Chime (523, 659, 784, 1047Hz)	Bill saved successfully
Error	Low buzz (200Hz + 150Hz)	Bill creation failed
Click	Short tick (800Hz)	Available for UI hooks

4.3 CSS Micro-interactions

- .active-elevate - scale(0.97) on button press
- .pb-safe - env(safe-area-inset-bottom) padding
- .bounce-in - keyframe animation (0.35s cubic-bezier)
- .cart-pulse - scale pulse animation (0.4s)
- .skeleton-shimmer - gradient shimmer animation (1.4s)

4.4 Settings Integration

- animationsEnabled - toggles Framer Motion page transitions
- soundsEnabled - toggles Web Audio API beeps
- Both persisted in settings store + localStorage
- Controlled from Bill Settings page

Files Created (Phase 4)

src/lib/animations.ts (Framer Motion variants)
src/lib/sounds.ts (Web Audio API sound effects)

Files Modified (Phase 4)

src/components/layout.tsx (page transitions)
src/pages/billing/index.tsx (sounds + cart badge + success screen)
src/stores/settings.ts (animationsEnabled, soundsEnabled)
src/index.css (micro-interaction utilities)

5. Files Created - Complete List

Phase	File	Purpose
P1	lib/db/migrations/001_gst_and_fixes.sql	Database migration for GST + indexes + fixes
P3	src/pages/settings/devices.tsx	Device Center - scanners, printers, tests
P3	src/pages/settings/bill-settings.tsx	Bill Settings - paper, footer, toggles
P3	src/pages/products/variants.tsx	Multi-unit variant creation UI
P4	src/lib/animations.ts	Framer Motion animation presets
P4	src/lib/sounds.ts	Web Audio API sound effects

6. Files Modified - Complete List

Phase	File	Change
P1/P2	lib/db/src/schema/products.ts	Added GST fields
P1	lib/db/src/schema/activity_log.ts	Added shop_id
P1	artifacts/api-server/src/routes/bills.ts	5 bug fixes + shop isolation
P1	artifacts/api-server/src/routes/products.ts	SQL-level filtering
P1	src/components/auth-guard.tsx	Added onboarding to public paths
P1	src/components/auth-provider.tsx	Added syncFromShop call
P1	src/pages/auth/onboarding.tsx	Redirect after creation
P1/P2	src/pages/billing/index.tsx	GST, safe-area, scanner, sounds
P2	src/pages/products/new.tsx	GST form fields
P2	src/pages/products/edit.tsx	GST form fields
P1/P2	src/stores/cart.ts	GST fields, clamp, shop key
P1	src/stores/settings.ts	syncFromShop, persistToServer
P2	src/lib/receipt.ts	Tax Invoice format
P1	src/pages/settings/store.tsx	Settings sync
P3	src/App.tsx	3 new routes
P3	src/pages/more/index.tsx	Menu items
P3	src/pages/products/detail.tsx	Variants button
P4	src/components/layout.tsx	Page transitions
P4	src/stores/settings.ts	animationsEnabled, soundsEnabled
P4	src/index.css	Micro-interaction utilities

7. Bugs Fixed - Complete Table

Bug	Phase	Status	Fix Summary
BUG-001	P1	Fixed	SELECT FOR UPDATE prevents stock race condition
BUG-002	P1	Fixed	Shop isolation on GET and CANCEL bill
BUG-003	P1	Fixed	Walk-in customer (0 and null)
BUG-004	P1	Fixed	Bill discount subtracts from profit
BUG-005	P1	Fixed	Crypto suffix prevents bill number collision
BUG-006	P1	Fixed	SQL WHERE filters instead of JS filtering
BUG-007	P1	Fixed	GET /bills shopId SQL filter
BUG-008	P1	Fixed	SELECT FOR UPDATE for all bill ops
BUG-009	P1	Fixed	shop_id on activity_log
BUG-010	P1	Fixed	Cart keyed by shopId
BUG-012	P1	Fixed	Onboarding in public paths
BUG-013	P1	Fixed	syncFromShop + persistToServer
BUG-016	P1	Fixed	Negative discount validation
BUG-017	P1	Fixed	Cart badge safe-area padding
BUG-018	P1	Fixed	0/null Walk-in handling
BUG-019	P1	Fixed	Onboarding redirect
BUG-020	P1	Fixed	Real stock check for bundles
BUG-021	P1	Fixed	Keyboard wedge scanner
BUG-022	P4	Fixed	SelectItem value="" crash -> value="walkin"

8. Features Added - Complete Table

Feature	Phase	Status
Stock race condition protection	P1	Done
Shop isolation (all GET/CANCEL/POST)	P1	Done
Unique bill numbers (crypto suffix)	P1	Done
Cart shop-keyed persistence	P1	Done
Settings server sync	P1	Done
HSN Code support	P2	Done
GST Rate (0/5/12/18/28%)	P2	Done
CGST/SGST/IGST breakdown	P2	Done
GST Inclusive/Exclusive toggle	P2	Done
Tax Invoice receipt format	P2	Done
Device Center	P3	Done
Bill Settings page	P3	Done
Multi-Unit Variants UI	P3	Done
Framer Motion page transitions	P4	Done
Sound effects (cart, scan, success, error)	P4	Done
Cart badge spring animation	P4	Done
Bill success checkmark animation	P4	Done
CSS micro-interactions	P4	Done
Sound/Animation toggles	P4	Done

9. Remaining Work / Future Ready

- Purchase Entry - Record when you buy stock from suppliers; auto-updates buy price and logs stock movement
- Bluetooth Scanner Deep Integration - Auto-detect paired BT scanner in Device Center
- Multi-Shop Support - Full CRUD for shops, user invites, role-based access
- Offline Mode - IndexedDB + service worker for offline billing
- Analytics Dashboard - Revenue trends, top products, margin analysis
- Supplier Management - Supplier directory, purchase history, payment tracking
- Employee/Staff Accounts - Role-based access (owner, cashier, manager)
- Backup & Export - CSV/Excel export for products, bills, customers
- Mobile App - PWA or native app for shopkeepers
- Customer Loyalty - Points system, repeat customer tracking

10. Final Status

Phases 1 to 4 - COMPLETED

19 bugs fixed | 6 new files | 21 files modified

GST billing is production-ready. Device Center and Bill Settings are functional.

Animations and sound effects are working. Ready for Phase 5 (Purchase Entry + Analytics).

